HIGH-RESOLUTION WIND AND EDDY COVARIANCE ANALYSIS OF HEAT AND MOMENTUM TRANSPORT IN CHICAGO

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ABSTRACT

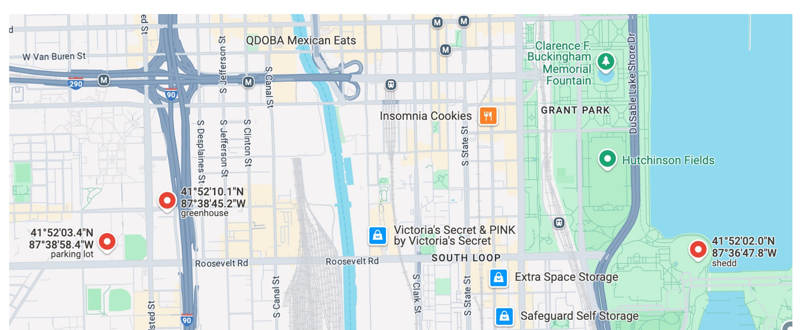
Urban environments challenge traditional eddy covariance (EC) methods due to their heterogeneous terrain and dynamic wind patterns.

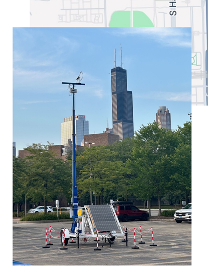
This study analyzes high-frequency, 3D wind data (July 27–28, 2024) from sonic anemometers across three Chicago sites—UIC Parking Lot, UIC Greenhouse, and Shedd Aquarium.

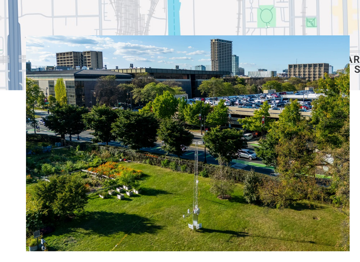
A novel approach using flexible averaging windows was applied to calculate sensible heat flux, momentum flux, and turbulent kinetic energy, addressing the limitations of conventional 30-minute stationarity assumptions.

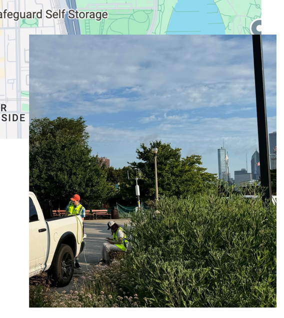
The results demonstrate that the calculated fluxes align with those obtained using EC software (EddyPro), validating the method's applicability to urban environments.

This research provides experts with a versatile framework for analyzing fluxes in complex urban terrains, offering an alternative to standard EC assumptions while maintaining consistency with established methods.



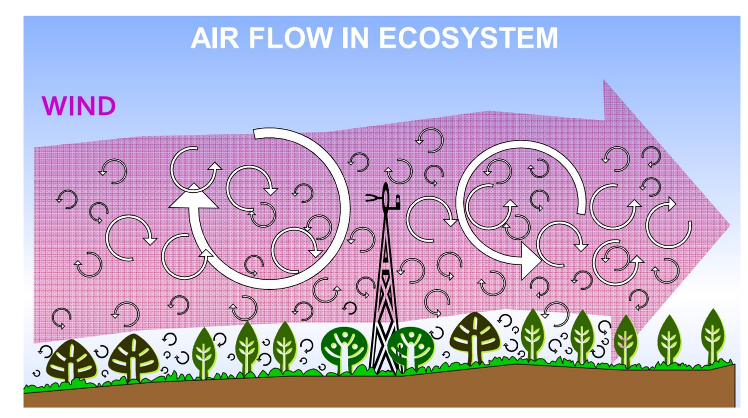






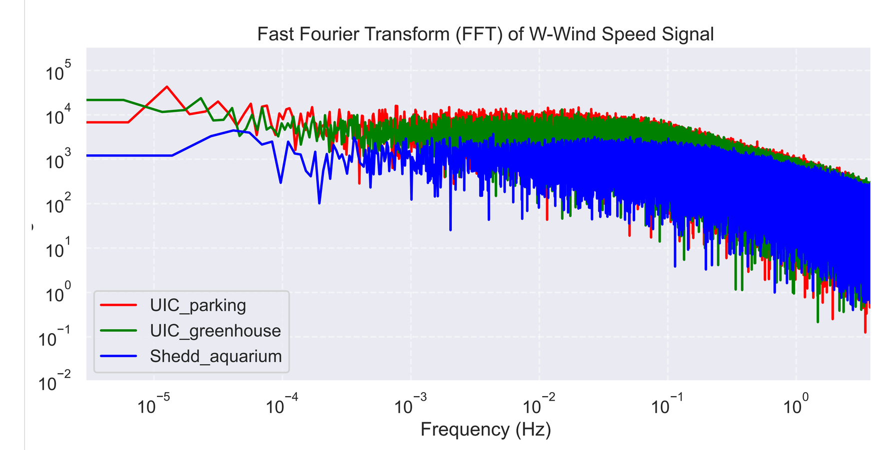
WIND EDDY

- A localized circular motion of air within a larger flow
- W-wind component plays crucial role in atmospheric vertical transport



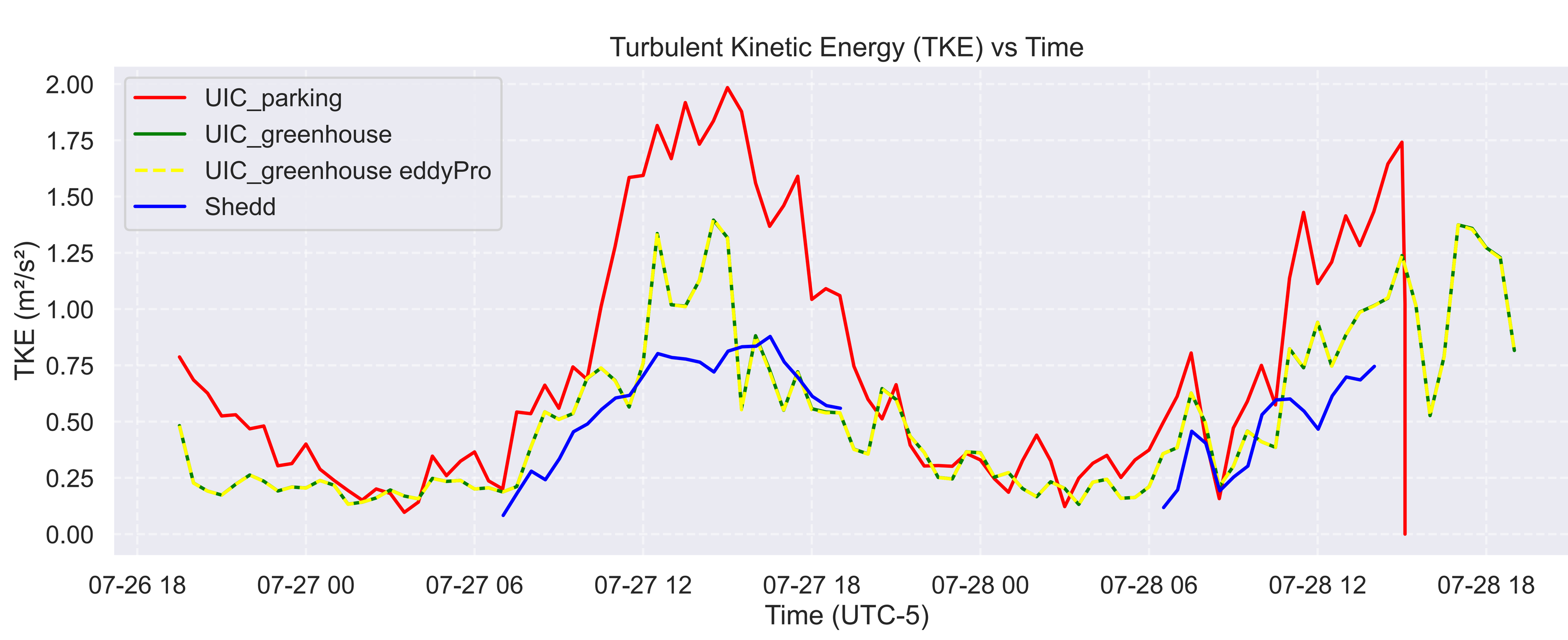
W-WIND

- Vertical wind component
- Demonstrates good turbulent flow
  - Kolmogorov's theory, -5/3 slope within inertial subrange
  - Energy concentrated at larger eddies and decreases as smaller eddies dominate
- Similar spectral signatures
  - Shedd FFT only incorporates daytime data

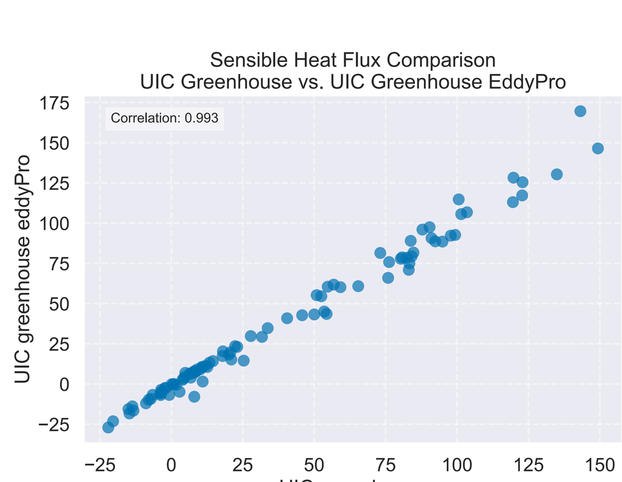


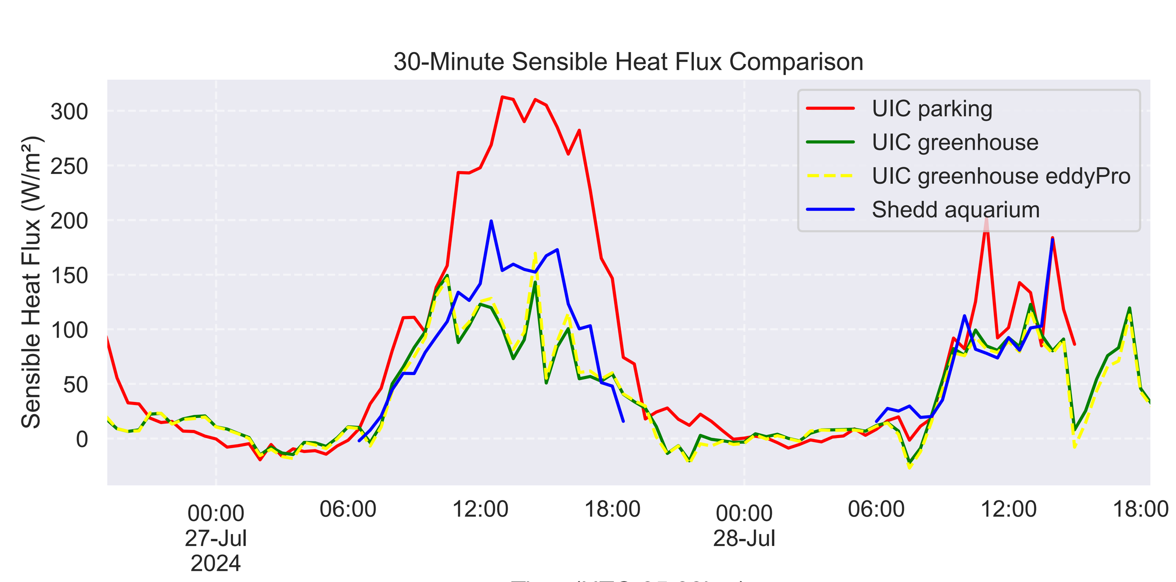
CREATING FUNCTIONS

- Ran a simple spike removal function (100-observations windows, all data over 3 window standard deviations are removed)
- Calculated TKE using Bodoni et al. 2019 $TKE = \frac{1}{2}(\sigma_u^2 + \sigma_v^2 + \sigma_w^2)$

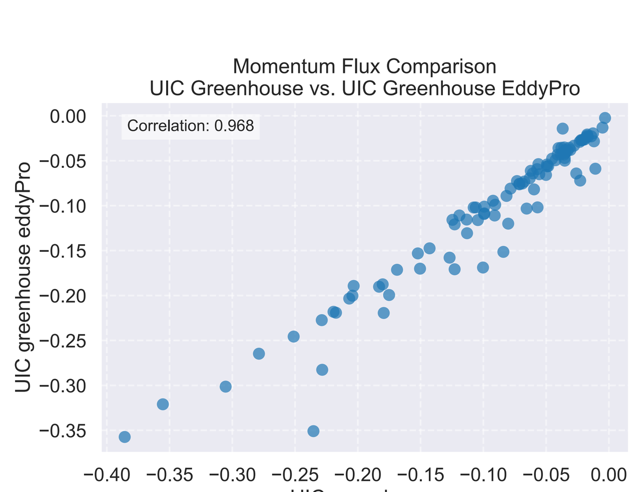


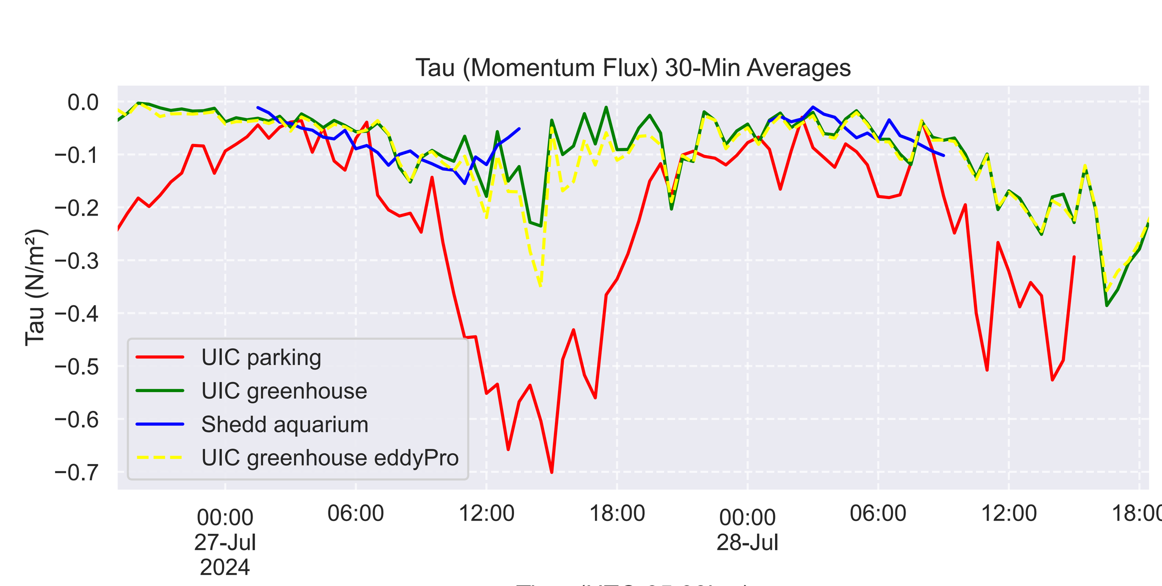
- Calculated Sensible Heat Flux using the Eddy Covariance Method $H_o = \rho_a c_p \overline{w'T_s'}$





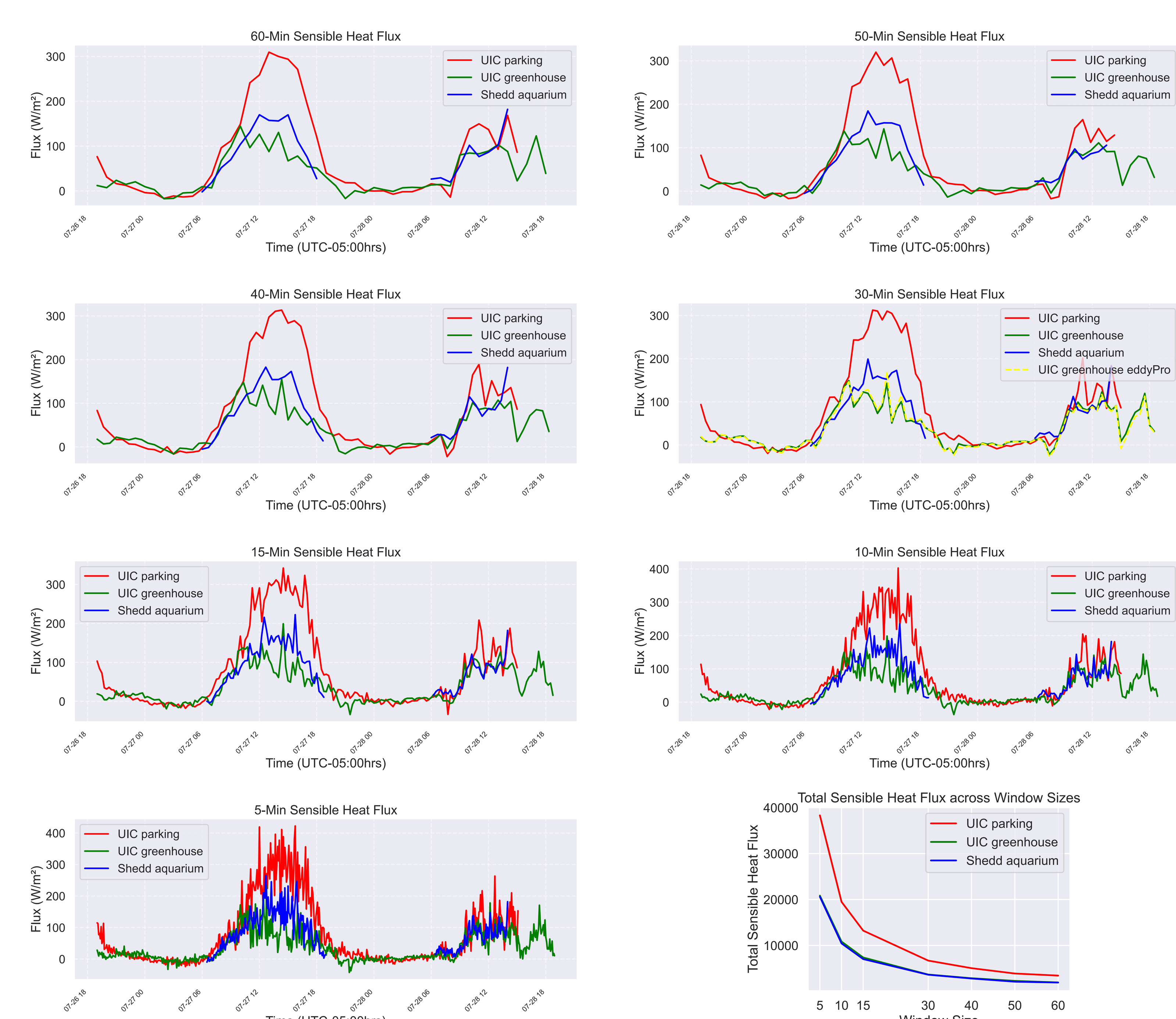
- Calculate Momentum Flux using Van Dijk et al. 2004 EC Method $T_0 = \rho_a \sqrt{\overline{u'w'}^2 + \overline{v'w'}^2}$





*Assumptions of P_a = 1.2 kg/m³ and C_p = 1005 J/kg

FLEXIBLE AVERAGING PERIODS



CONCLUSION

- A novel approach using flexible window durations overcomes limitations of traditional 30-minute averaging in urban flux calculations
- Fluxes align with EC software (EddyPro), confirming the method's reliability for complex urban terrains

NEXT STEPS

- Apply stricter stationarity criteria to enable accurate fluxes over non-standard averaging intervals
- Automate stationarity filtering to support flexible, high-resolution eddy covariance analysis

DELIVERABLES

- Package with basic data cleaning and flexible-window calculations for sensible heat, momentum flux, and TKE
  jpiot1/flux_calculations
- Sensible heat flux, momentum flux, and TKE datasets for CROCUS-Urban IOP (July 27-28)

REFERENCES

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- Pal S ; Raut B ; Muradyan P ; Tuftedal M ; O'Brien J ; Sullivan R ; Grover M ; Jackson R ; Berkelhammer M ; Wawrzyniak E ; Gonzalez-Meler M ; Cho A ; McNicol G ; Sankaran R ; Shahkarami S ; Collis S (2024): CROCUS High-Frequency Measurements of CO₂, H₂O, Wind, and Temperature at University of Illinois Chicago. Community Research on Climate and Urban Science Urban Integrated Field Laboratory (CROCUS UIFL), ESS-DIVE repository. Dataset. doi:10.15485/2473255 accessed via https://data.ess-dive.lbl.gov/datasets/doi:10.15485/2473255 on 2025-07-17.